Supplemental Material
Gate-dependent vacancy diffusion in graphene

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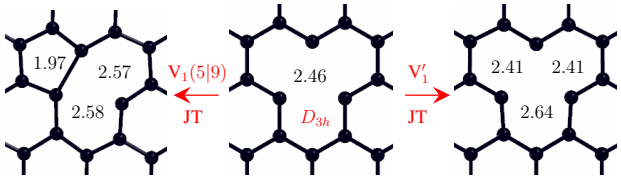

Supercell	Vacancy structure	E_f (eV)	ΔE_f (meV)
6×4 $(\sqrt{3},3)a_0$; 96 atoms	V$_1$(5|9)	7.66	0
	V$'_1$	7.73	74
10×5 $(\sqrt{3},3)a_0$; 200 atoms	V$_1$(5|9)	7.64	0
	V$'_1$	7.83	190
13×8 $(\sqrt{3},3)a_0$; 416 atoms	V$_1$(5|9)	7.56	0
	V$'_1$	7.79	232

FIG. 1. The symmetric D_{3h} vacancy goes through different Jahn-Teller (JT) distortions and lower energy. The bond lengths are shown in Å. The variation in formation energy E_f and the difference ΔE_f between V$_1$(5|9) and V$'_1$ is shown for different sized supercells. The E_f is calculated as $E_f = 1/n\left(E_{\text{vac}} - \frac{N-n}{N}E_{\text{pristine}}\right)$, where n is the number of missing atom, and N is the total number of atoms in the supercell.

A monovacancy in graphene lattice is formed in the absence of a single C-atom during growth or under irradiation, and results in three dangling σ bonds. Subsequent relaxation of the under coordinated atoms, which form a equilateral 2.46 Å vacancy structure, leads to different vacancy structures. There are two possible ways in which an equilateral triangle with D_{3h} symmetry could distort into isosceles C_{2v} triangle due to Jahn-Teller effect. In the Jahn-Teller distorted ground state, the two C-atoms form a weak σ bond while the dangling bond survives at the apical atom. The resultant vacancy structure thus consists of 5-9 ring V$_1$(5|9), where the C-atoms at the pentagon are 1.97 Å apart and form an isosceles triangle with the apical atom that is 2.58 Å away. The other JT distorted metastable structure V$'_1$ is found to be 0.19 eV higher, which has two shorter (2.41 Å) and one longer C−C distance. Different JT relaxation mechanisms and the respective formation energies are shown in the Figure 1 for varied supercell sizes. The formation energy E_f is calculated to be $\sim$ 7.65 eV and does not show a strong dependence on the supercell size.

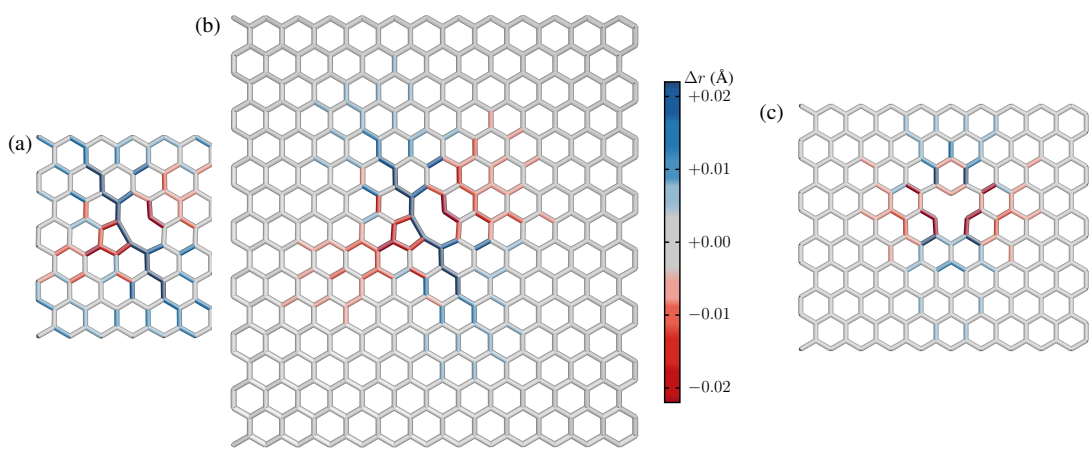

FIG. 2. The defect induced strain field for V$_1$(5|9) vacancy structure calculated with (a) 6×4$(\sqrt{3},3)a_0$ and (b) 13×8$(\sqrt{3},3)a_0$ supercells. (c) The same for the V$'_1$ vacancy structure calculated with a 10×5$(\sqrt{3},3)a_0$ supercell. The strain field is calculated as the difference in the bond length Δr with the defect-free lattice.

The electron doped case is more relevant in the context of dynamic JT reconstruction than the hole doped situation as the charge is transferred from the TEM grid

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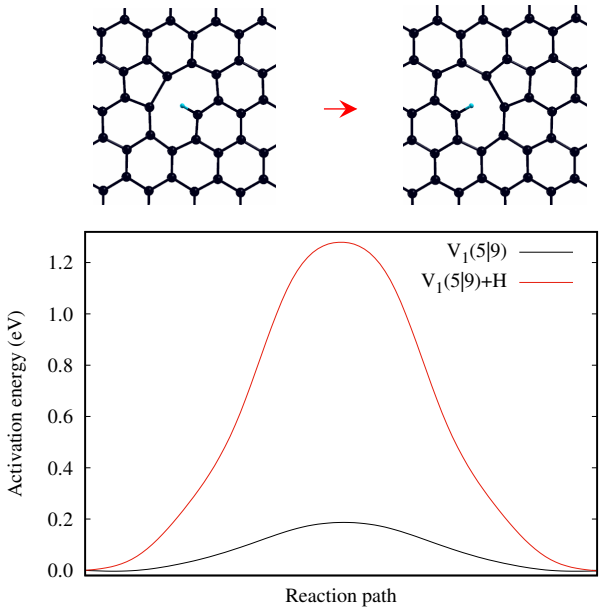

FIG. 3. Calculated activation barrier for the dynamic JT distortion for the hydrogenated vacancy, $V_1(5|9)$+H $\rightarrow$ 120°-$V_1(5|9)$+H. The activation barrier is found to be much larger compared to the same for bare $V_1(5|9)$ vacancy.

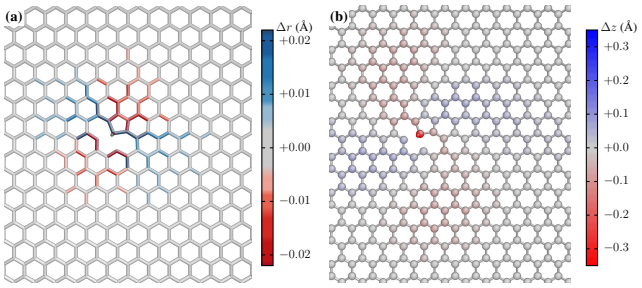

FIG. 4. The in-plane (a) and out-of-plane (b) lattice relaxation at the transition state for the hydrogenated vacancy as shown for the $13\times8(\sqrt{3},3)a_0$ supercell.The transition state structure is completely different from the case for the bare vacancy. The in-plane lattice deformation is much localized, and the out-of-plane relaxation is much smaller. The maximum Δz is found to be 0.09 Å, compared to 0.34 Åfor the bare vacancy case.

to graphene. However, one could still discuss the effect of hole doping where the barrier reduces to 0.16 eV for 1.89×10^{13} cm^{-2} density that corresponds to a time-scale of 10^{-3} s at 77 K.

Vacancy migration is anticipated to be influenced by the strain field that is generated by the defect itself, which trigger defect-defect interaction. Thus, the determination of the correct supercell size becomes crucial to study an individual vacancy defect. We find that the strain field produced by the $V_1(5|9)$ defect to be long-range in nature, and is extended over 2 nm distance from the defect center [Figure 2(a) and (b)]. Thus, while a small $6\times4(\sqrt{3},3)a_0$ supercell is inadequate, a much bigger $13\times8(\sqrt{3},3)a_0$ supercell is determined to be redundant. In this regard, the $10\times5(\sqrt{3},3)a_0$ supercell is sufficient with negligible defect-defect interaction. Compared to the $V_1(5|9)$ defect structure, the strain field generated by the V_1' vacancy structure is much localized around the vacancy [Figure 2(c)].

TABLE I. Crossover temperature for quantum tunnelling calculated for different doping level.

Doping level (cm^{-2})	ν^* (cm^{-1})	T^* (K)
3.77×10^{13} e	376.44	86.20
1.89×10^{13} e	361.68	82.82
Neutral	323.83	74.15
1.89×10^{13} h	312.79	71.62
3.77×10^{13} h	309.39	70.84

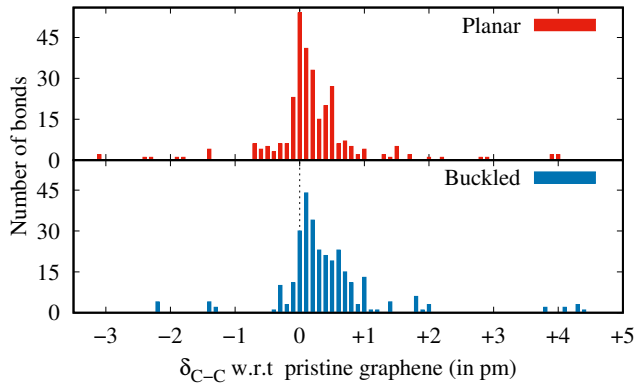

FIG. 5. Distribution of C–C bonds for the (a) planar and (b) buckled transition states are shown for the $10\times5(\sqrt{3},3)a_0$ supercell. The difference in bond length is calculated with respect to the defect-free graphene. The out-of-place lattice relaxation is important for overall strain relaxation during migration.

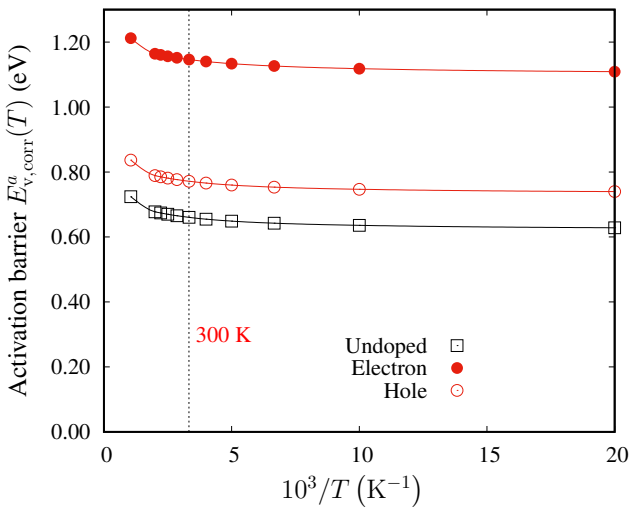

FIG. 6. Quantum mechanical Wigner correction to the classical activation barrier for the $V_1(5|9)$ vacancy, $E^a_{\mathrm{v,corr}}(T) = E^a_{\mathrm{v}} - \delta E_{\mathrm{v,wig}}(T)$ shown for the neutral and doped cases with 1.89×10^{13} cm^{-2} density. $E^a_{\mathrm{v,corr}}(T)$ converges to the classical barrier E^a_{v} at a high temperature, while at very low temperature, the zero-point energy correction is sufficient. The quantum tunnelling correction is not included above 100 K.

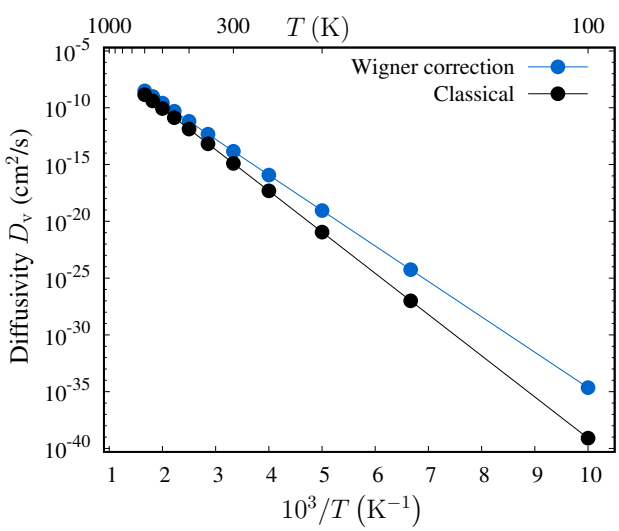

FIG. 7. Diffusivity calculated using the Wigner corrected barrier for T >100 K and compared with the diffusivity calculated with the classical barrier. At T =300 K, the Wigner corrected diffusivity is one order of magnitude higher than the classical one.